



Addis Ababa University
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SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !



Addis Ababa Institute of Technology

School of Mechanical and Industrial Engineering

Engineering Materials I

(MEng 2122)

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April, 2020

Chapter - 4

Imperfection in Solids

Basic Contents

- Types of imperfections
- Point defect
- Linear defect
- Surface/interfacial/ defect
- Volume defect
- Grain size determination
- Microscopic examination of microstructures

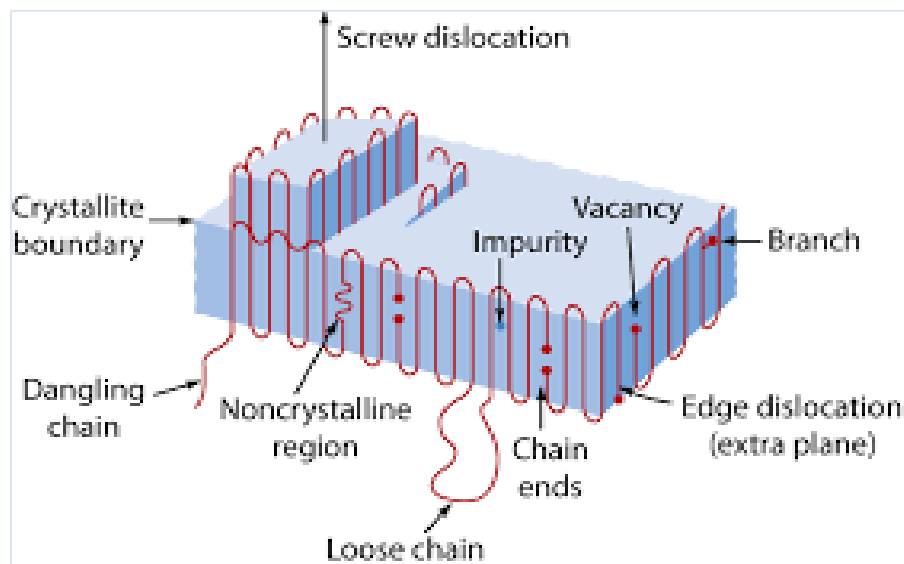
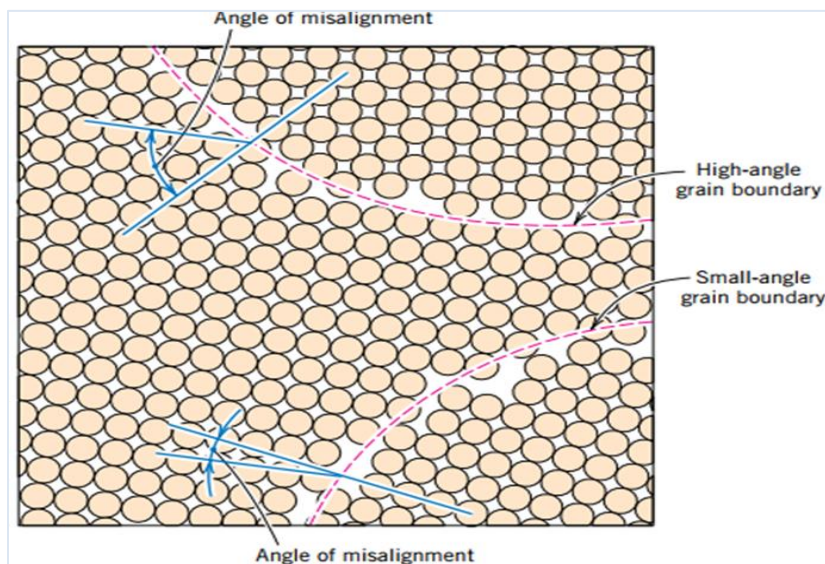
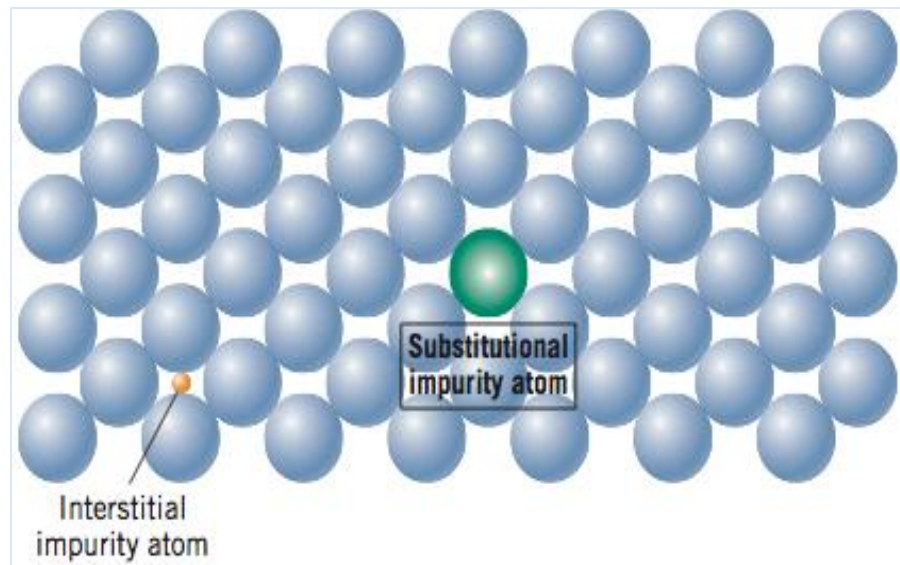
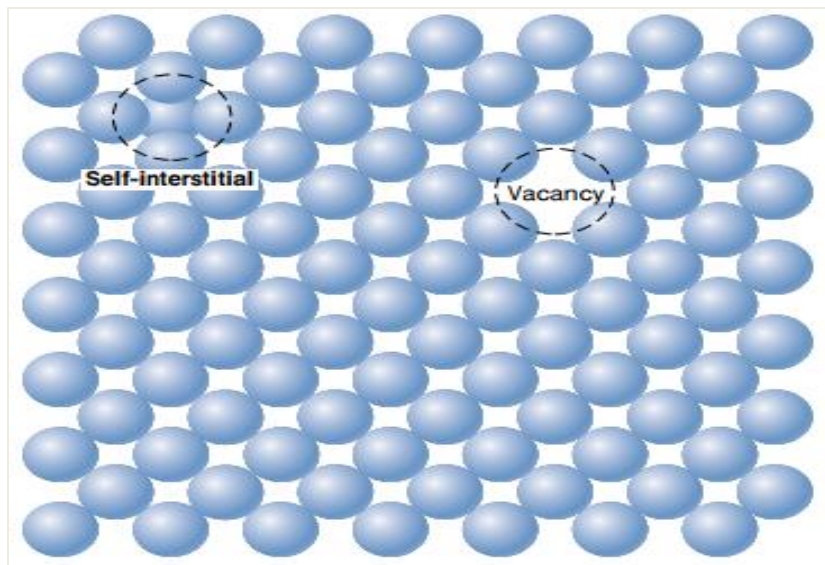
Chapter - Learning Objectives

After completing this chapter, you should be able to:

- Describe both vacancy and self-interstitial crystalline defects.
- Calculate the equilibrium number of vacancies in a material at some specified temperature.
- List and explain the two types of solid solutions.
- Calculate the weight percent and atom percent for each element, with given masses and atomic weights of two or more elements in a metal alloy.
- Describe the type of dislocations (edge, screw, and mixed) and show their illustration, dislocation line and the direction of dislocation extension.
- Describe the atomic structure within the vicinity of a grain boundary and a twin boundary.

Imperfection in Solids

Overview of Imperfections



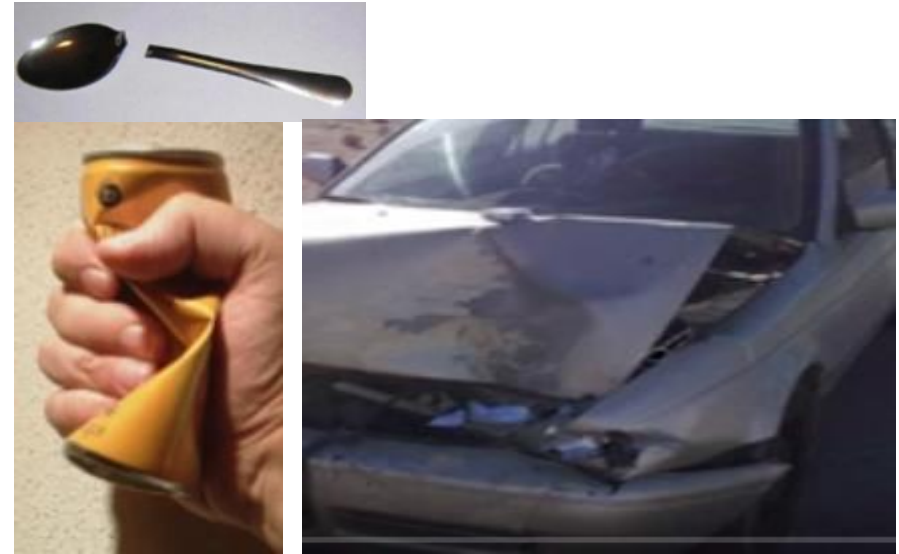
Introduction to Imperfections in Solid Materials

- In real materials we find: with **crystalline defects** or **lattice irregularity**. These abnormalities are termed as **imperfection**.
- A **crystalline defect** refers to a lattice irregularity having one or more of its dimensions on the order of an atomic diameter.
- Most real materials have one or more “**errors in perfection**” with dimensions on the order of an atomic diameter to many lattice sites.
- Classification of **crystalline imperfections** is according to:
 - 1) **geometry** (point, line or plane), or
 - 2) **dimensions** of the defect.

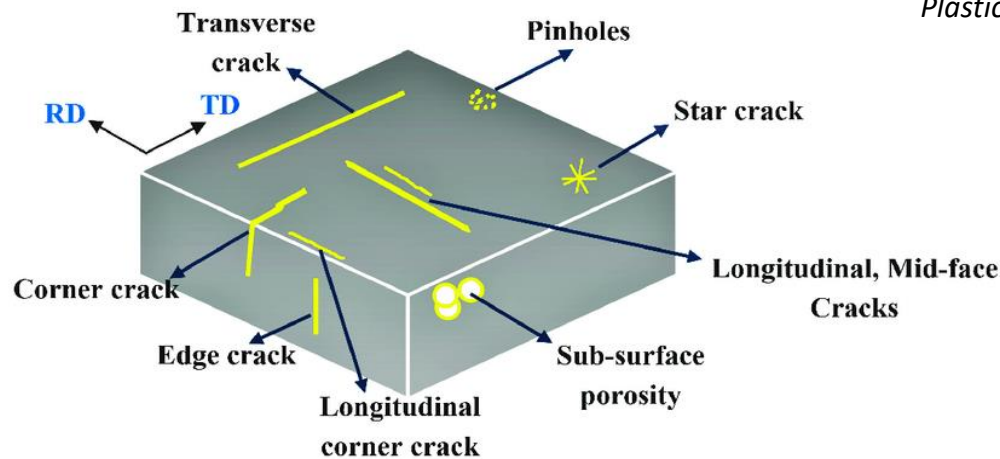
Imperfections - Plastic Deformation

Dislocations and plastic deformation

- *Plastic deformation* is permanent deformation.
- Plastic deformation occurs when dislocations move along planes in metals.



Plastic deformation (in visible manner)



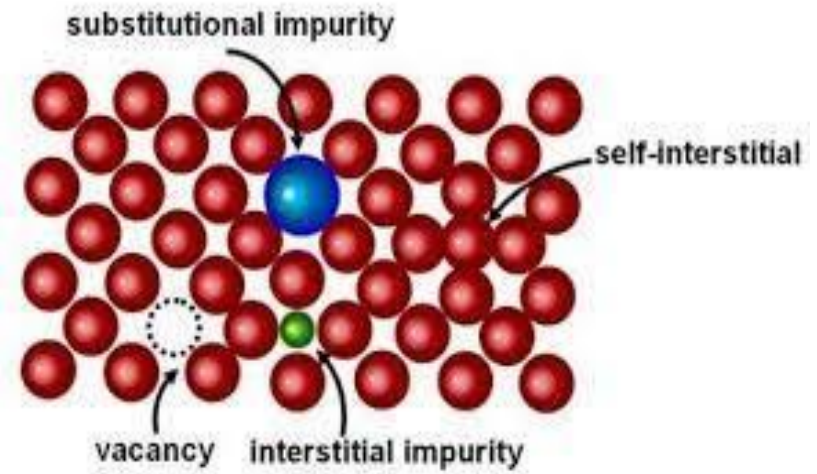
General imperfections

Types of Imperfections

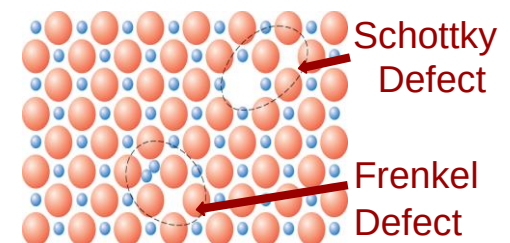
- Types of imperfections:
 - 1) Point defects – associated with one/two atomic positions
 - 2) Linear defects – one-dimensional
 - 3) Surface /interfacial/ defects – boundaries - two dimensional
 - 4) Volume defects – three dimensional.
- Impurities in solids are also considered, because impurity atoms may exist as point defects.

Point Defects

- Types of point defects:
 - Vacancies
 - Interstitial impurities
 - Substitutional impurities
 - Schottky defect
 - Frenkel defect

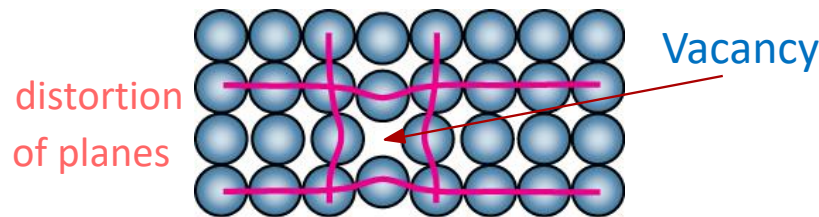


- **Schottky defect** is occurred in those crystals in which sizes of ions is almost similar. - a paired set of cation and anion vacancies.
- **Frenkel defect** is occurred in those ionic crystals in which sizes of ions (anions and cations) show large differences. - a cation is out of place.

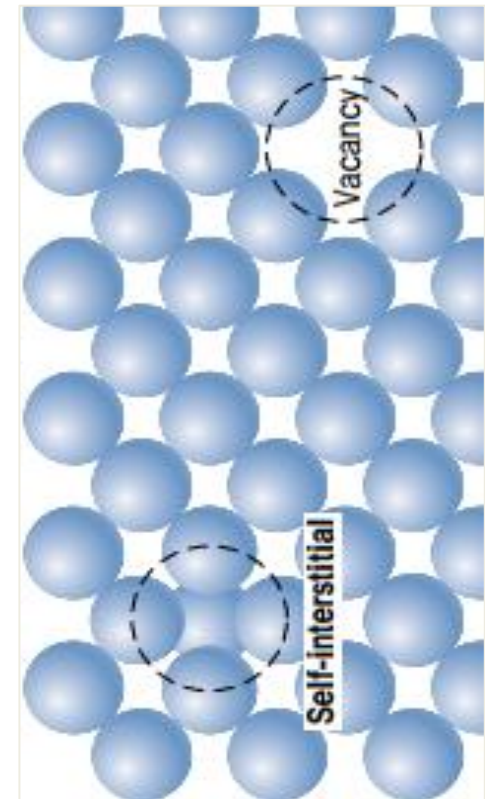
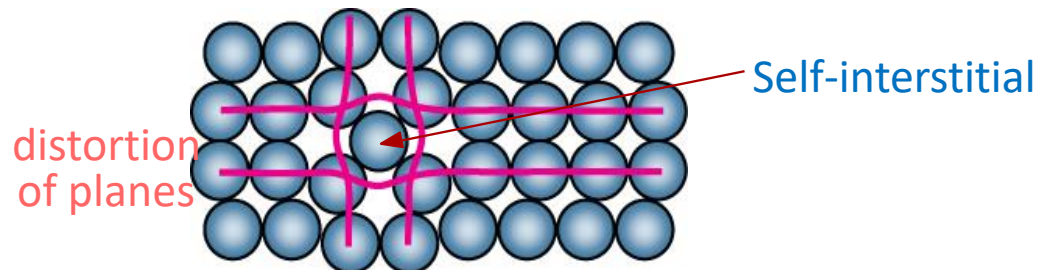


Point Defects

- The simplest of the point defects is a **vacancy**, or vacant lattice site.
- All **crystalline solids** contain vacancies. It is not possible to create such a material that is free of these defects.
- **Vacancies:**
 - vacant atomic sites in a structure.



- **Self-Interstitials:**
 - "extra" atoms positioned between atomic sites.



Point Defects

- A **self-interstitial** is an atom from the crystal that is crowded into an *interstitial site*.
- Principles of **thermodynamics** is used to explain the necessity of the existence of vacancies in crystalline solids.
- The presence of vacancies **increases** the entropy (randomness) of the crystal.
- The **equilibrium number** of vacancies for a given quantity of material depends on and increases with temperature as follows: (an Arrhenius model):

$$N_v = N \exp(-Q_v/kT)$$

Equilibrium no. of vacancies → N_v *Total no. of atomic sites* → N *Energy required to form vacancy* → Q_v

T = absolute temperature in Kelvin
 k = gas or Boltzmann's constant

Point Defects

Impurity in Alloys

- A pure metal consisting of only one type of atom just isn't possible.
- Impurity or foreign atoms are always present, and some exist as crystalline point defects.
- Most familiar metals are not highly pure; rather, they are alloys, in which impurity atoms have been added intentionally to impart specific characteristics to the material.
- Alloying is used in metals to improve mechanical properties such as strength and corrosion resistance.
- Alloying in metals refers to point defect due to formation of solid solution.

Point Defects

- The addition of impurity atoms to a metal results in the formation of a solid solution or a new *second phase*.
- The solid solution formation is depending on:
 - kinds of impurity,
 - their concentrations, and
 - the temperature of the alloy.
- A solid solution (alloy) consists of a solvent and solute.
 - *Solvent* is the element or compound that is present in the greatest amount. Solvent atoms are also called *host atoms*.
 - *Solute* is used to denote an element or compound present in a minor concentration.

Point Defects

- A **solid solution** is also compositionally homogeneous; the impurity atoms are randomly and uniformly dispersed within the solid.
- **Impurity** point defects are found in solid solutions, of which there are two types: **substitutional** and **interstitial**.

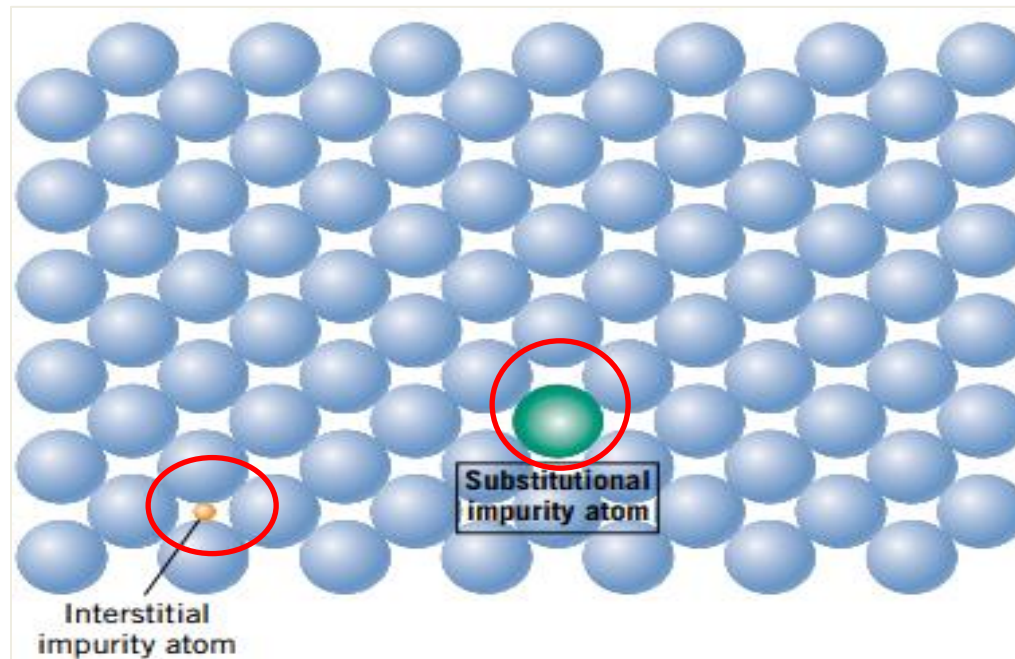


Figure: Substitutional and interstitial impurity atoms.

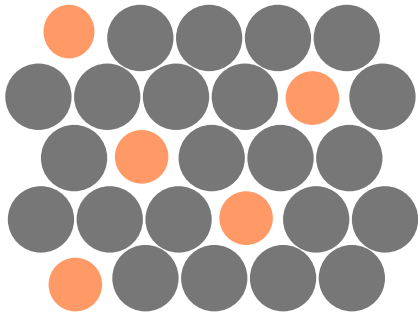
Point Defects

- For the *substitutional type*, *solute* or impurity atoms replace or *substitute* for the host atoms.
- Determining features of the *solute* and *solvent* atoms to which the former dissolves in the latter. These are expressed as four *Hume–Rothery rules*, as follows:
 1. *Atomic size factor*: appreciable quantities of solute present - only when the difference in atomic radii between the two atom types is less than 15%.
 2. *Crystal structure*: for appreciable solid solubility, the crystal structures for metals of both atom types must be the *same*.
 3. *Electronegativity factor*: if both elements are the more electropositive – better to form intermetallic compound rather than substitutional solid solution.
 4. *Valences*: a metal has a tendency to dissolve another metal of higher valency than lower valency.

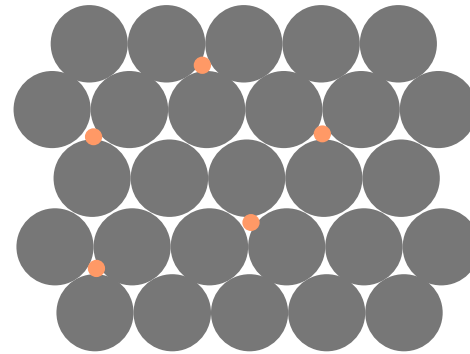
Point Defects

Two outcomes if impurity (B) added to host (A):

- Solid solution of B in A - random destination of point defects.



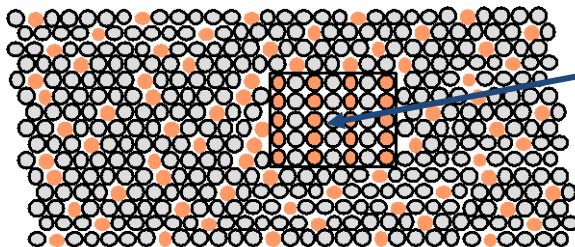
OR



Substitutional solid solution.
(e.g., Cu in Ni)

Interstitial solid soln.
(e.g., C in Fe)

- Solid solution of B in A - plus particles of a new phase (usually for a larger amount of B).



Second phase particle
- different composition
- often different structure.

Point Defects

Specification of Composition

- The **composition** or *concentration* of an alloy is necessary to express in terms of its constituent elements.
- The two most common ways to **specify composition** are **weight** (or mass) percent and **atom** percent.
- The basis for **weight percent** (wt%) is the weight of a particular element relative to the total alloy weight.

Computation of **weight percent**
(for a two-element alloy);

$$C_1 = \frac{m_1}{m_1 + m_2} \times 100$$

$m_{1,2}$ = mass of
component 1 &
2 respectively.

Computation of **atom percent**
(for a two-element alloy);

$$C_1' = \frac{n_{m1}}{n_{m1} + n_{m2}} \times 100$$

$n_{m1,2}$ = number of moles of component 1 & 2

$$n_{m1} = \frac{m_1'}{A_1}$$

Point Defects

■ Composition Conversions: (Wt% and At%)

$$C_1' = \frac{C_1 * A_2}{C_1 * A_2 + C_2 * A_1} \times 100$$

$$C_2' = \frac{C_2 * A_1}{C_1 * A_2 + C_2 * A_1} \times 100$$

*Converts from wt%
to At% (A_i is atomic
weight)*

$$C_1 = \frac{C_1' * A_1}{C_1' * A_1 + C_2' * A_2} \times 100$$

$$C_2 = \frac{C_2' * A_2}{C_1' * A_1 + C_2' * A_2} \times 100$$

*Converts from at%
to Wt% (A_i is atomic
weight)*

■ Example:

1. What is the composition in atom percent, of an alloy that consists of 92.5wt% Ag and 7.5wt% Cu? (Hint: Atomic weight of Cu = 63.55g/mol and atomic weight of Ag = 107.87g/mol)

$$\text{Ans. } C_{Ag} = \frac{C_{Ag} * A_{Cu}}{(C_{Ag} * A_{Cu}) + (C_{Cu} * A_{Ag})} * 100 = \frac{92.5 * 63.55}{(92.5 * 63.55) + (7.5 * 107.87)} * 100 = 87.902 \text{ at\%}$$

$$C_{Cu} = \frac{C_{Cu} * A_{Ag}}{(C_{Ag} * A_{Cu}) + (C_{Cu} * A_{Ag})} * 100 = \frac{7.5 * 107.87}{(92.5 * 63.55) + (7.5 * 107.87)} * 100$$

$$= 12.098 \text{ at\%}$$

Point Defects

$$C_1 = \frac{m_1}{m_1 + m_2 + m_3 + \cdots + m_n} \times 100$$

$$C_1 + C_2 = 100$$

$$C'_1 + C'_2 = 100$$

$$C'_1 = \frac{n_{m1}}{n_{m1} + n_{m2} + n_{m3} + \cdots + n_{mn}} \times 100$$

Conversion of weight percent to mass per unit volume (for a two-element alloy)

$$C''_1 = \left(\frac{C_1}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}} \right) \times 10^3$$

$$C''_2 = \left(\frac{C_2}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}} \right) \times 10^3$$

Computation of density (for a two element metal alloy)

$$\rho_{\text{ave}} = \frac{100}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}}$$

$$\rho_{\text{ave}} = \frac{C'_1 A_1 + C'_2 A_2}{\frac{C'_1 A_1}{\rho_1} + \frac{C'_2 A_2}{\rho_2}}$$

Computation of atomic weight (for a two-element metal alloy)

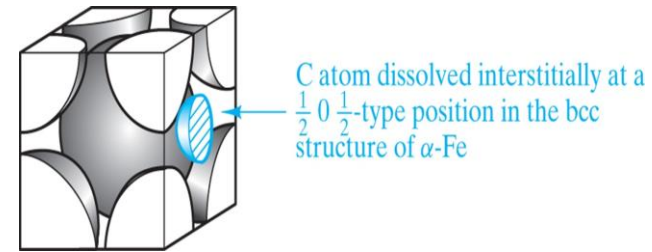
$$A_{\text{ave}} = \frac{100}{\frac{C_1}{A_1} + \frac{C_2}{A_2}}$$

$$A_{\text{ave}} = \frac{C'_1 A_1 + C'_2 A_2}{100}$$

Point Defects

- **Interstitial solid solution** applies to carbon in α -iron.

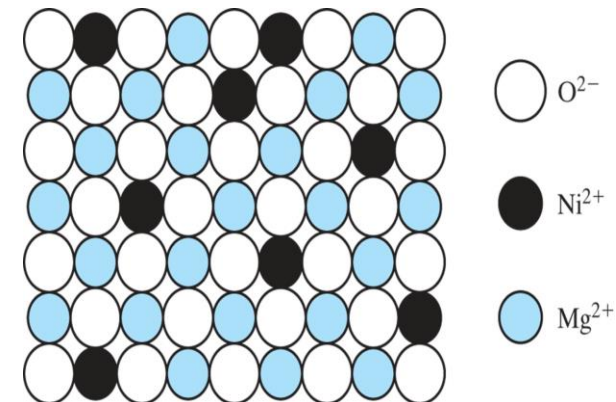
The carbon atom is small enough to fit with some strain in the interstice (or opening) among adjacent Fe atoms in this important steel structure.



But the interstitial solubility is quite low since the size mismatch of the site to the radius of a carbon atom is only about 1/4

- Random, **substitutional solid** solution can occur in Ionic Crystalline materials as well. Here of NiO in MgO.

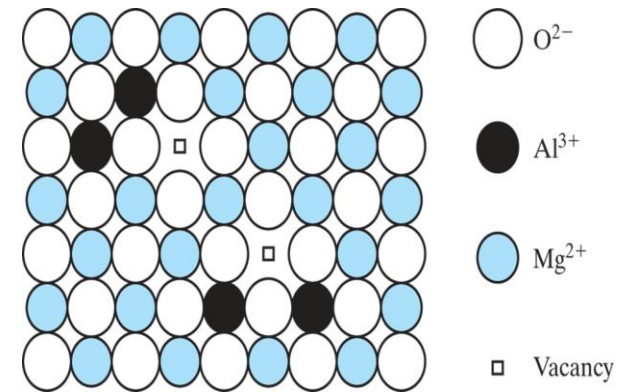
The O^{2-} arrangement is unaffected. The substitution occurs among Ni^{2+} and Mg^{2+} ions.



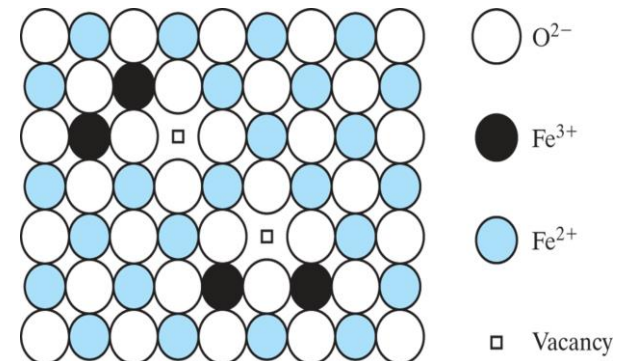
Point Defects

- A *substitution solid* solution of Al_2O_3 in MgO is not as simple as the case of NiO in MgO .

The requirement of *charge neutrality* in the overall compound permits only two Al^{3+} ions to fill every three Mg^{2+} vacant sites, leaving one Mg^{2+} vacancy.



- Iron oxide, Fe_{1-x}O with $x \approx 0.05$, is an example of a nonstoichiometric compound. Similar to the case of Figure 4.6, both Fe^{2+} and Fe^{3+} ions occupy the cation sites, with one Fe^{2+} vacancy occurring for every two Fe^{3+} ions present.



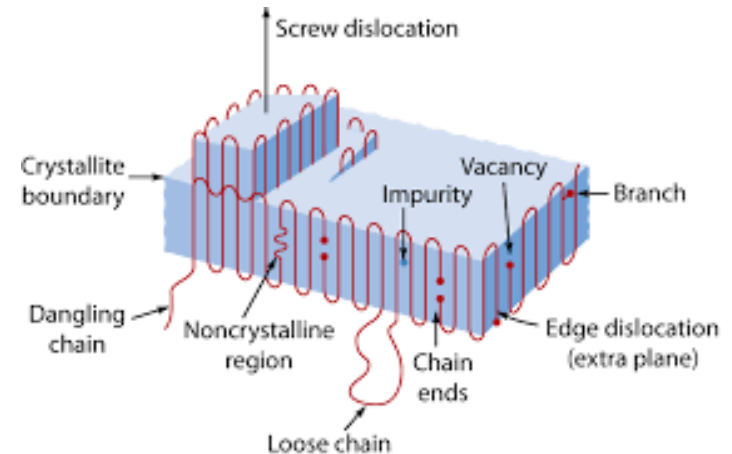
- Equilibrium concentration of defects $\sim e^{-Q_D / kT}$

Linear Defects

- Linear defects are called **dislocations**.
- A **dislocation** is a linear or one-dimensional defect around which some of the atoms are misaligned.

- Types of **dislocations**:

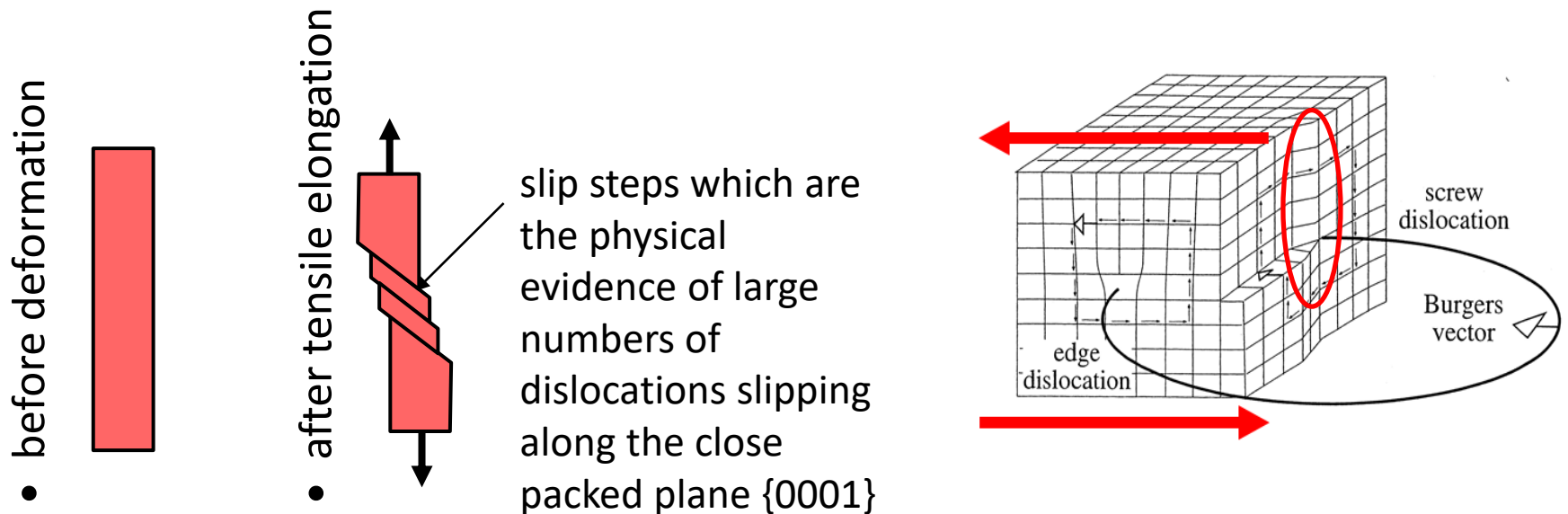
- 1) Edge dislocation
- 2) Line dislocation
- 3) Screw dislocation



- **Edge dislocation**: an extra portion of a plane of atoms, or half-plane, the edge of which terminates within the crystal.
- **Line dislocation**: it is a linear defect that centers on the line that is defined along the end of the extra half-plane of atoms.

Linear Defects

- **Screw dislocation**: is formed by a shear stress that is applied to produce the distortion.
 - slip between crystal planes result when dislocations move,
 - this motion produces permanent (plastic) deformation.
- Schematic of Zinc (HCP):



Linear Defects

Linear Defects (Dislocations)

- Are one-dimensional defects around which atoms are misaligned.
- Edge dislocation:
 - extra half-plane of atoms inserted in a crystal structure.
 - **b (the Berger's vector)** is $\perp$ (perpendicular) to dislocation line.

Burger's vector, b : is a measure of lattice distortion and is measured as a distance along the close packed directions in the lattice.

- The **magnitude** and **direction** of the lattice distortion associated with a dislocation are expressed in terms of a **Burgers vector**, denoted by **b**. (see Figure 4.4)

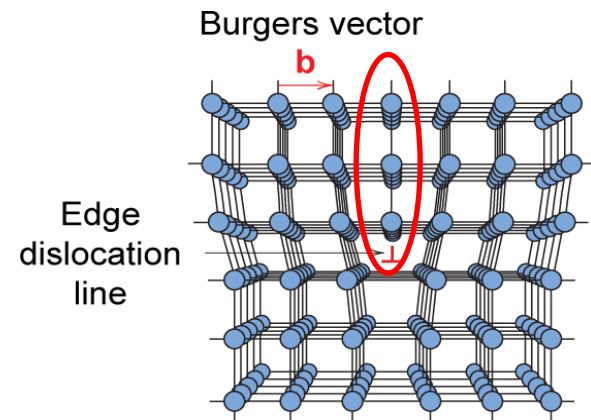
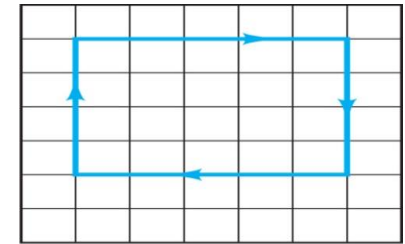


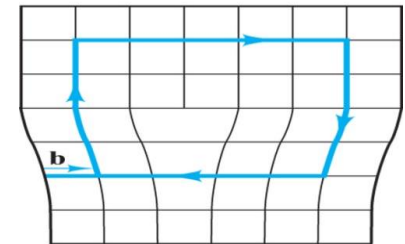
Figure 4.4 The atom positions around an edge dislocation; extra half-plane of atoms shown in perspective. (Callister 9th ed.)

Linear Defects

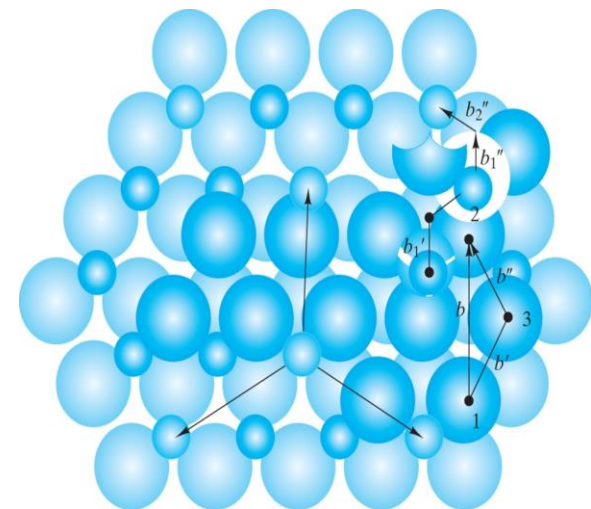
- Definition of the **Burgers vector** $\mathbf{b}$, relative to an **edge dislocation**. (a) In the perfect crystal, an $m \times n$ atomic step loop closes at the starting point. (b) In the region of a dislocation, the same loop does not close, and the closure vector $\mathbf{b}$ represents the magnitude of the structural defect. For the **edge dislocation**, the Burgers vector is perpendicular to the dislocation line.
- **Burgers vector** for the aluminum oxide structure. The large repeat distance in this relatively complex structure causes the Burgers vector to be broken up into two (for O^{2-}) or four (for Al^{3+}) partial dislocations, each representing a smaller slip step. This **complexity** is associated with the brittleness of ceramics compared with metals.



(a)



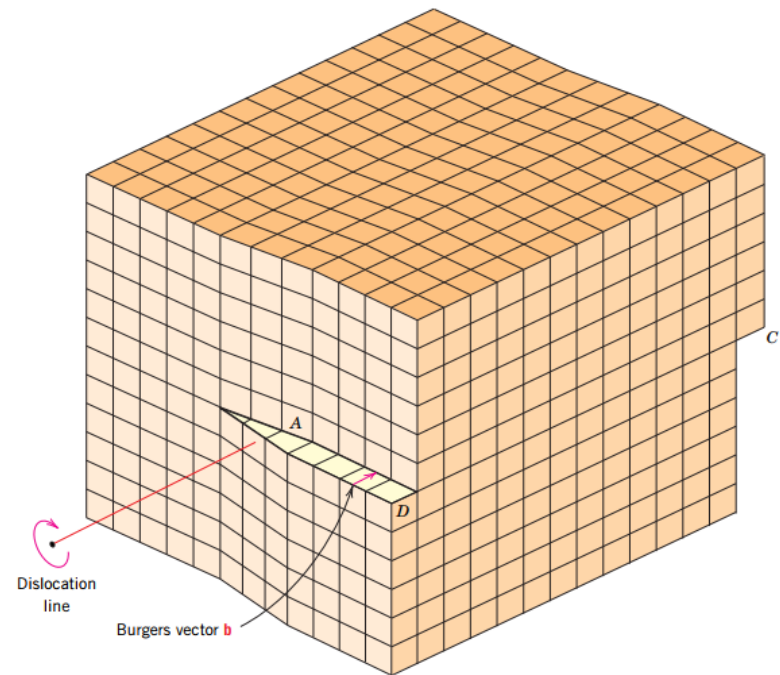
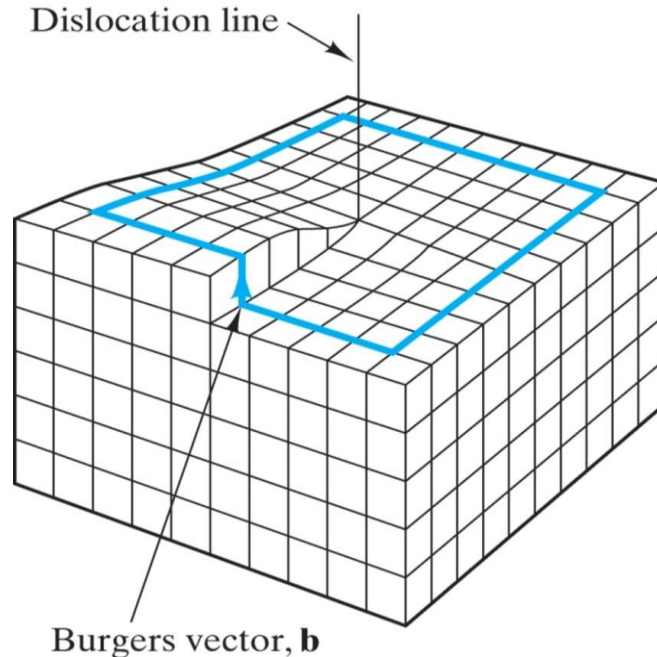
(b)



(W. D. Kingery, Introduction to Ceramics, 2nd ed.)

Linear Defects

- *Screw dislocation*: the spiral stacking of crystal planes leads to the Burgers vector being parallel to the dislocation line.
 - spiral planar ramp resulting from shear deformation.
 - $\mathbf{b}$ is $\parallel$ (parallel) to dislocation line.

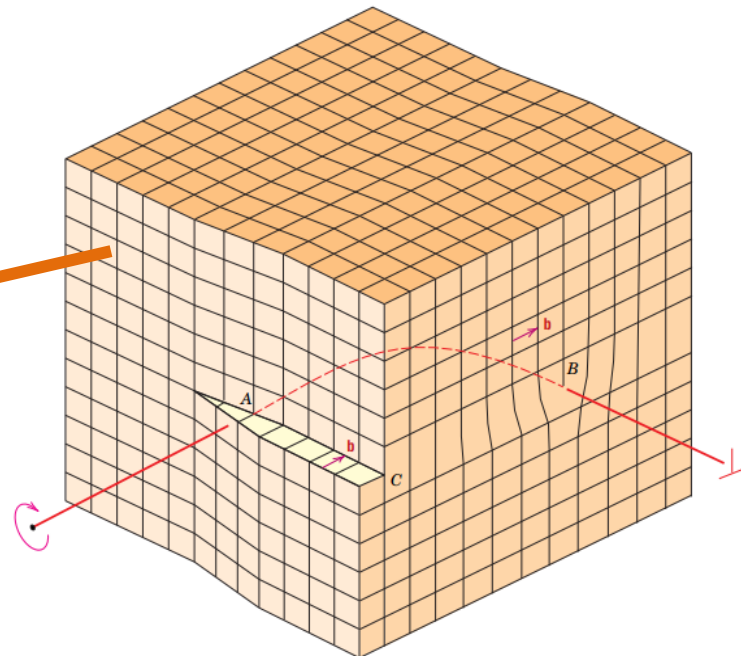
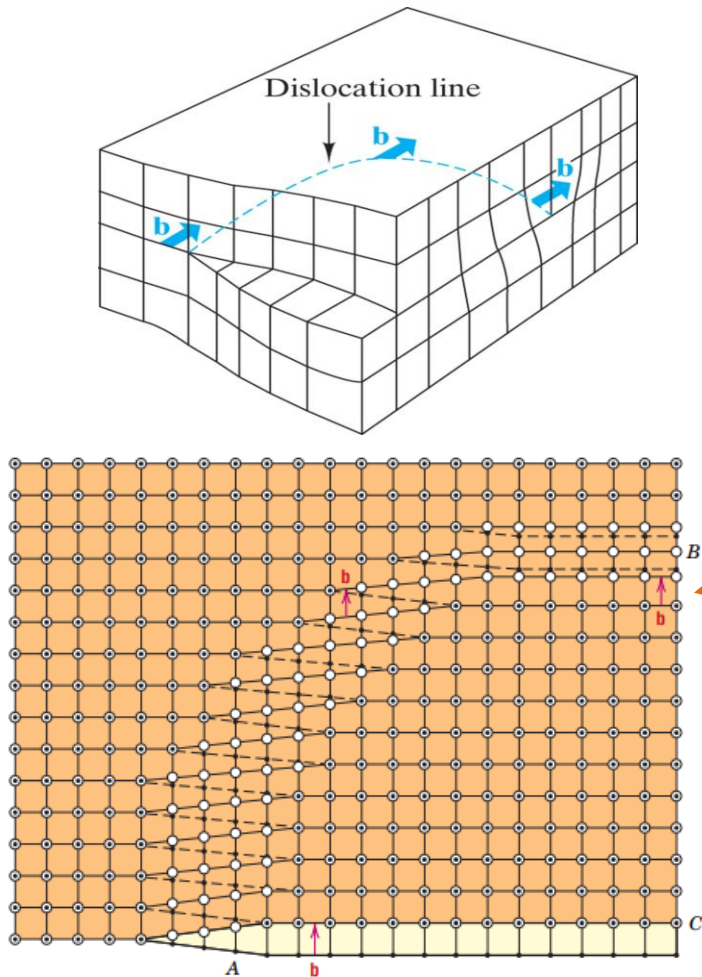


A screw dislocation within a crystal.

Linear Defects

- *Mixed dislocation*: this dislocation has both edge and screw character with a single Burgers vector consistent with the pure edge and pure screw regions.

An *edge dislocation* therefore moves in the direction of the Burgers vector, whereas a *screw dislocation* moves in a direction perpendicular to the Burgers vector.



A dislocation that has edge, screw, and mixed character.

Linear Defects

- Dislocations can be observed in crystalline materials using **electron-microscopic** techniques.
- A high magnification transmission **electron micrograph**, the *dark lines* are the *dislocations*.

Dislocations
are visible in
(T) electron
micrographs

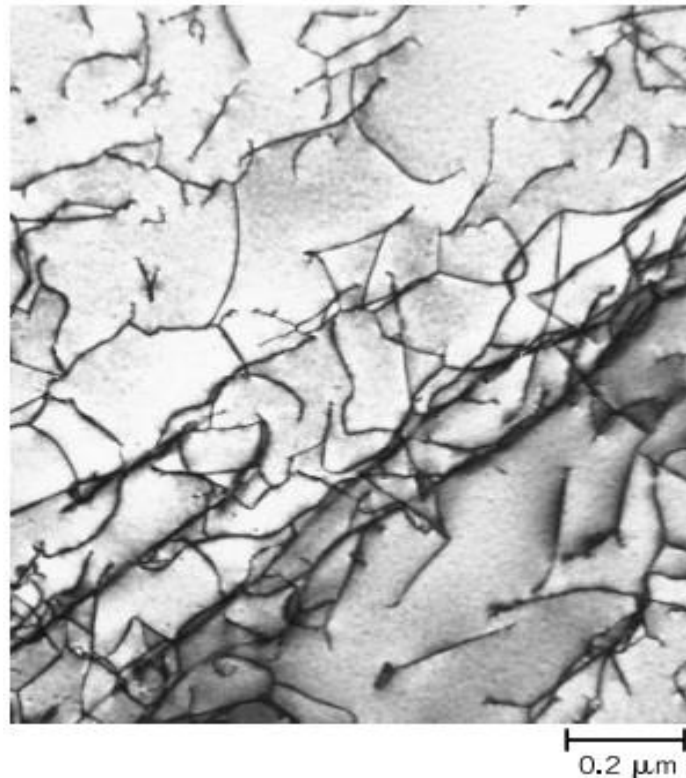


Figure 4.7: A transmission electron micrograph of a titanium alloy in which the dark lines are dislocations, 50,000x.

Surface Defects

- **Surface defects:** are also called **interfacial defects**, which are boundaries that *have two dimensions* and normally separate regions of the materials that *have different crystal structures* and/or crystallographic orientations.
- These **imperfections** include:
 - external surfaces,
 - grain boundaries,
 - phase boundaries,
 - twin boundaries, and
 - stacking faults.

Surface Defects

External Surfaces

- It is the most obvious boundary along which the **crystal structure** terminates.
- **Surface atoms** are not bonded to the maximum number of nearest neighbors. They are in a higher energy state than the atoms at interior positions.
- The **bonds** of these surface atoms that are not satisfied give rise to a surface energy, expressed in units of energy per unit area (J/m^2 or erg/cm^2).
- To reduce this energy, **materials tend to minimize**, if at all possible, the total surface area.

Surface Defects

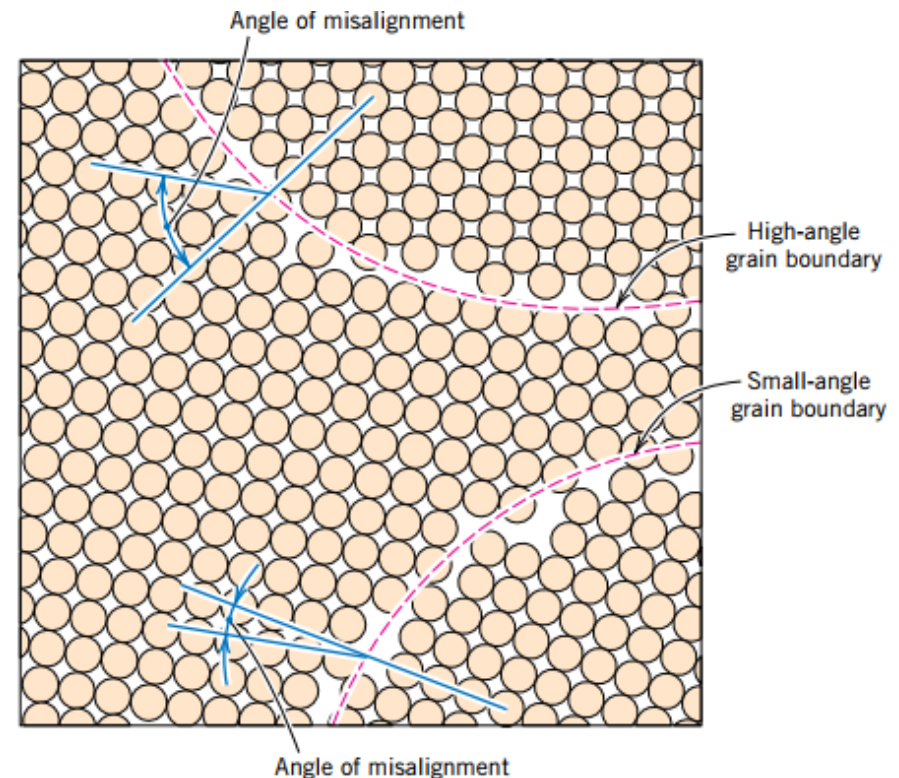
Grain Boundaries

- **Grain boundary** is the boundary separation of two small grains or crystals having different crystallographic orientations in polycrystalline materials.
- A **grain boundary** is represented schematically from an atomic perspective.
- Various degrees of **crystallographic misalignment** between adjacent grains are possible.
- When the orientation mismatch is slight, then it refers to *small (low-) angle grain boundary*.
- One simple **small-angle grain boundary** is formed when edge dislocations are *aligned in tilted* form. It is called a *tilt boundary*.

Surface Defects

Grain Boundaries

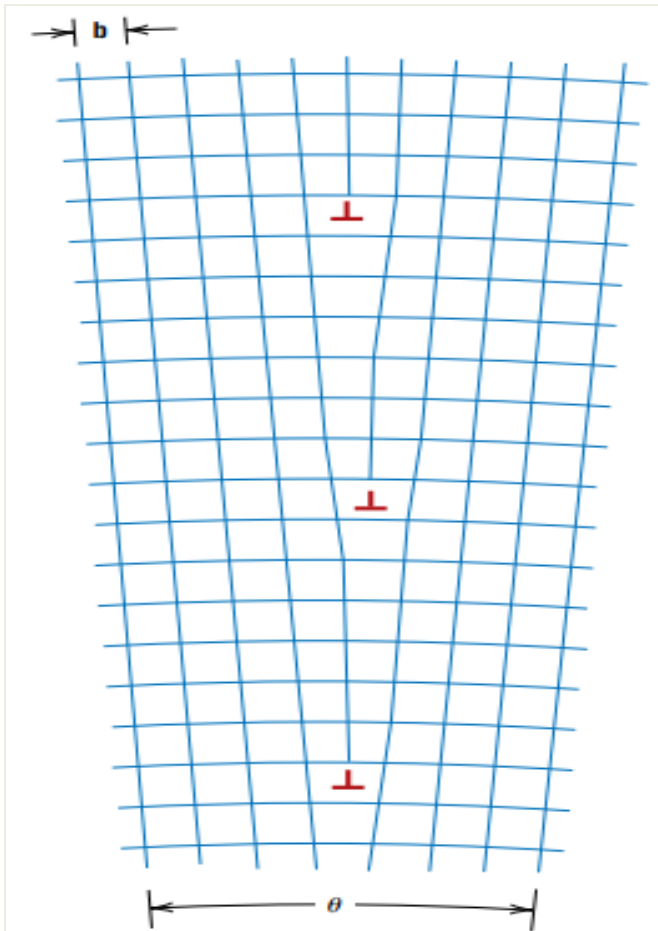
- When the **angle of misorientation** is *parallel* to the boundary, a *twist boundary* results.
- The **atoms** are bonded less regularly along a grain boundary.
- The magnitude of this energy is a function of the degree of misorientation.
- **Grain boundaries** are more chemically reactive than the grains themselves as a consequence of this boundary energy.



Shows small & high-angle grain boundaries and the adjacent atom positions.

Surface Defects

- A tilt boundary having an angle of misorientation results from an alignment of edge dislocations.



- The total interfacial energy is lower in large or coarse-grained materials than in fine-grained ones.

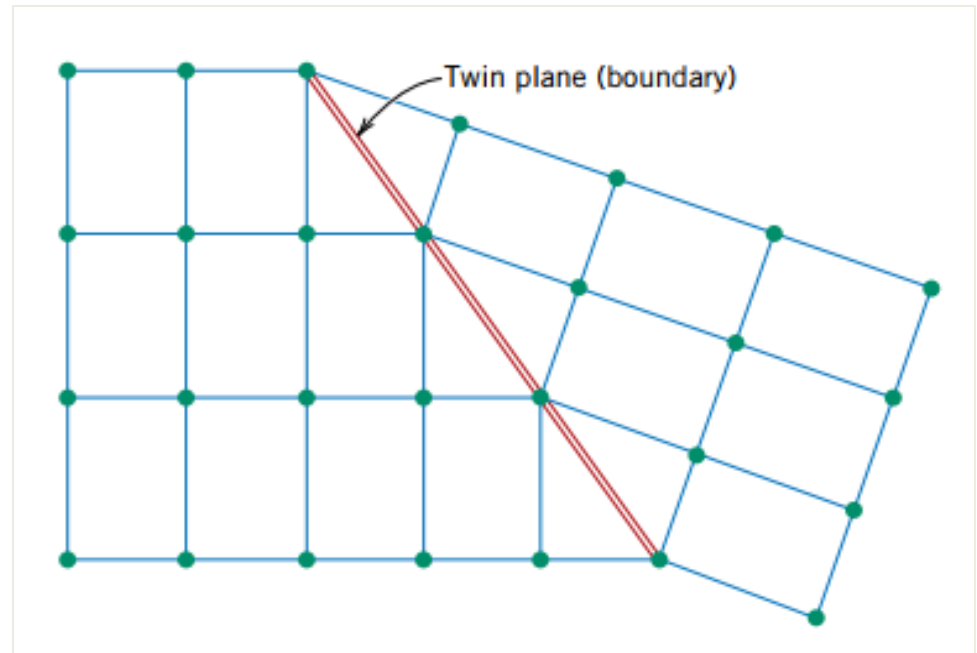


Figure 4.10: A twin plane or boundary and the adjacent atom positions (colored circles).

Surface Defects

Phase Boundaries

- *Phase boundaries* exist in multiphase materials, in which a different phase exists on each side of the boundary.
- Each of the constituent phases has its own distinctive **physical** and/or **chemical** characteristics.
- **Phase boundaries** play an important role in determining the mechanical characteristics of some multiphase metal alloys.

Twin Boundaries

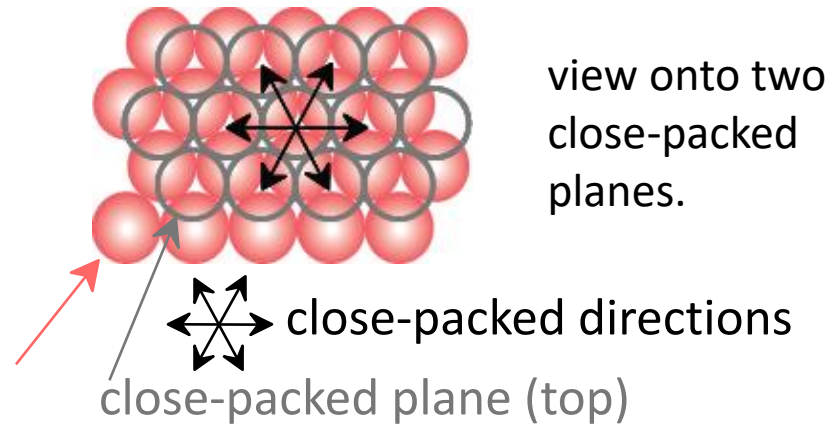
- A *twin boundary* is a special type of grain boundary across which there is a specific mirror lattice symmetry.
- **Atoms** on one side of the boundary are located in mirror-image positions to those of the atoms on the other side. (Figure 4.10)

Surface Defects

Dislocations and Crystal Structures

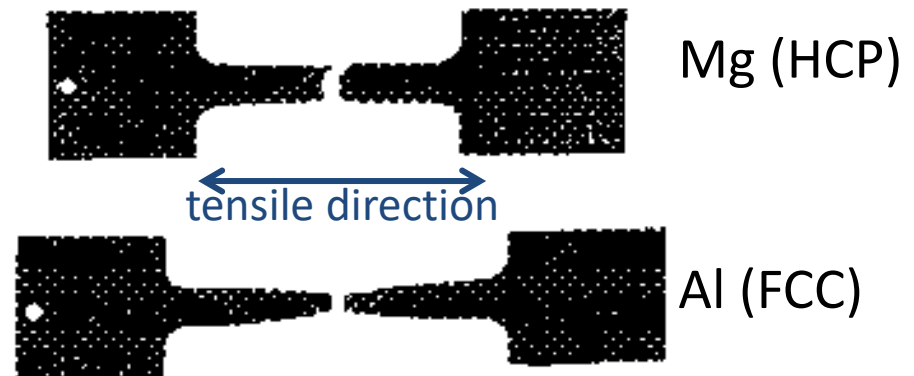
- Structure: **close-packed** planes & directions are preferred.

close-packed plane (bottom)



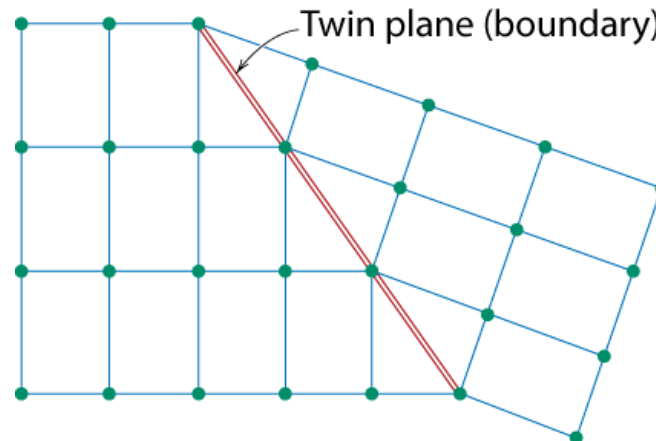
- Comparison among crystal structures:
FCC: many close-packed planes/directions;
HCP: only one plane, 3 directions;
BCC: none “super-close” many “near close”

- Specimens that were tensile tested.



Surface Defects

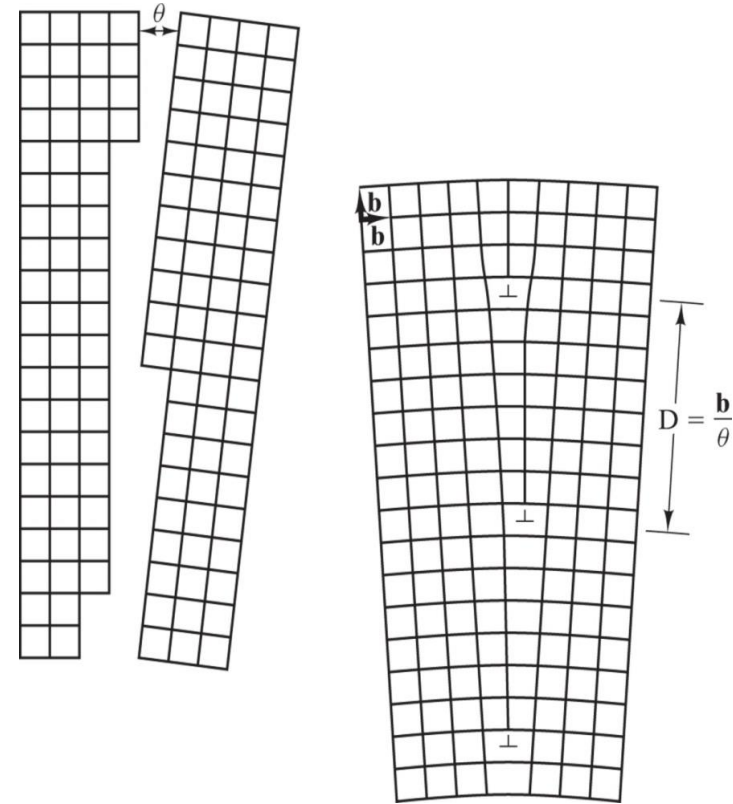
- **Annealing twins** are typically found in metals that have the FCC crystal structure. whereas
- **Mechanical twins** are observed in BCC and HCP metals.
- One case is a **twin boundary (plane)**
 - Essentially a reflection of atom positions across the **twinning plane**.



- **Stacking faults:**
 - Are in FCC metals an error in ABCABC packing sequence
 - Ex: ABCABABC

Surface Defects

- Simple *grain-boundary structure*. This is termed a *tilt boundary* because it is formed when two adjacent crystalline grains are tilted relative to each other by a few degrees (ϑ). The resulting structure is equivalent to isolated edge dislocations separated by the distance b/ϑ , where b is the length of the Burgers vector, **b**. (From W. T. Read, Dislocations in Crystals, McGraw-Hill Book Company, New York, 1953.)



Volume Defects

- Volume defects are also called bulk defects.
- Volume defects includes:
 - pores,
 - cracks,
 - foreign inclusions, and other phases.
- These defects are normally introduced during processing and fabrication steps.

Voids – coalesced vacancies

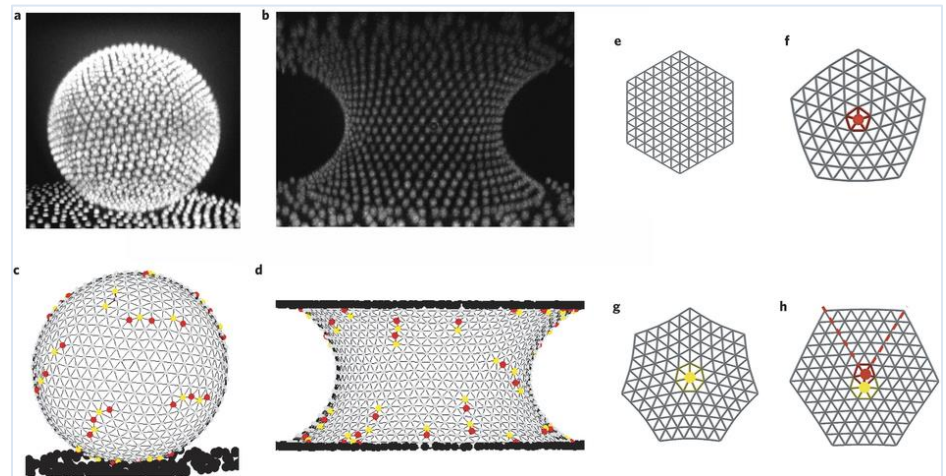
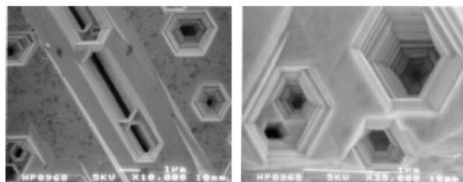
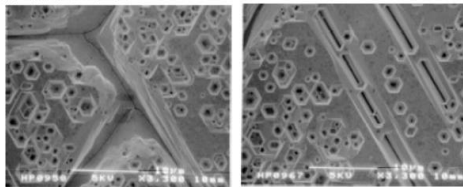
Cracks

Pits

Grooves

Inclusions

Precipitates



Volume Defects

Atomic Vibration

- Every atom in a solid material is **vibrating** very rapidly about its lattice position within the crystal.
- These **atomic vibrations** may be considered as defects.
- Not all atoms **vibrate** at the same frequency and amplitude or with the same energy.
- The **temperature(T°)** of a solid is really just a measure of the average **vibrational activity** of atoms and molecules.
- At room T° , a typical **vibrational frequency** is on the order of 10^{13} vibrations per second.
- Many **properties** and **processes** in solids are manifestations of this vibrational atomic motion.

Microscopic Examination of Defects

- It is important to *examine* the structural elements and defects that influence the properties of materials.
- Some structural elements are of *macroscopic dimensions*.
- For example, the *shape* and *average size* or diameter of the grains for a polycrystalline specimen are important structural characteristics.
- *Macroscopic grains* are often evident on aluminum streetlight posts and also on highway guardrails.
- *Grain size* and *shape* are only two features of what is termed the **microstructure**.

Microscopic Examination

- The commonly used microscopes in **microscopy** are:
 - 1) **optical**,
 - 2) **electron**, and
 - 3) **scanning probe microscopes**.
- These instruments aid in **investigations** of the microstructural features of all material types.
- Some of these **techniques** employ **photographic** equipment in conjunction with the **microscope**.
- The **photograph** on which the image is recorded is called a **photomicrograph**.
- **Microscopic examination** is an extremely useful tool in the study and ***characterization of materials***.



Microscopic Examination

- Applications of **microstructural examinations** are:
 - to ensure that the associations between the properties and structure (and defects) are properly understood,
 - to predict the properties of materials once these relationships have been established,
 - to design alloys with new property combinations,
 - to determine whether a material has been correctly heat-treated, and
 - to ascertain the mode of mechanical fracture.

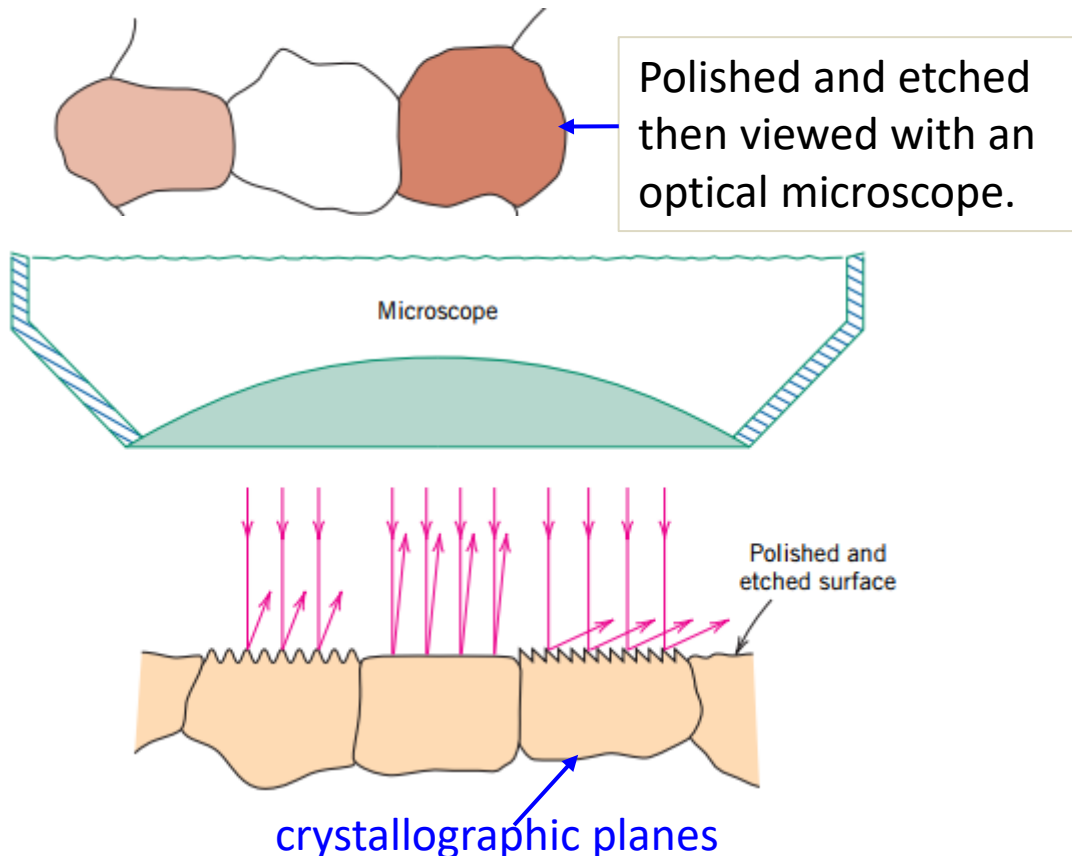


- Cross-section of a cylindrical copper ingot. The small, needle-shape grains may be observed, which extend from the center radially outward.

Microscopic Techniques

Optical Microscopy

- Polishing removes surface features (e.g., scratches)
- Etching changes reflectance, depending on crystal orientation since different crystallographic planes have different reactivity.



Etching characteristics and resulting surface texture vary from grain to grain.



Photomicrograph of a polycrystalline brass specimen, 60x.

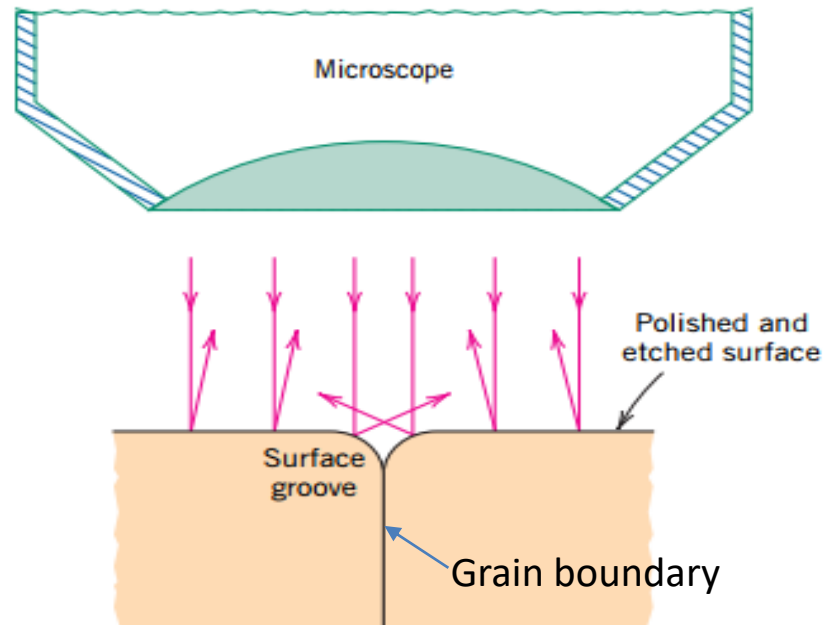
Microscopic Techniques

Optical Microscopy

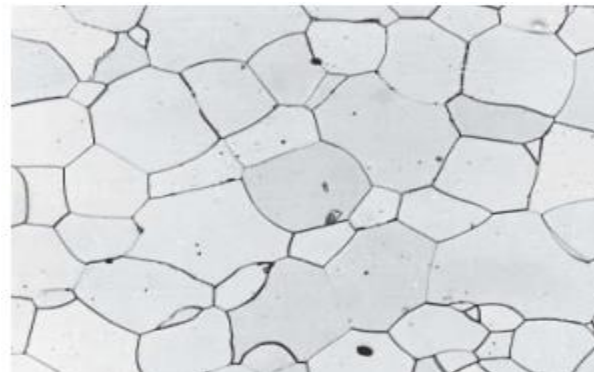
- An optical microscope's magnification capacity is up to 2000x.

Since Grain boundaries;

- are planar imperfections,
- are more susceptible to etching,
- may be revealed as dark lines,
- relate change in crystal orientation across boundary.



A grain boundary and its surface groove produced by etching.



Photomicrograph of the surface of a polished and etched Fe-Cr alloy.

Microscopic Techniques

Electron Microscopy

- An **electron microscope** is capable of much higher magnifications than an optical microscope (2000x). That is, $\geq 2000x$
- An image of the structure under investigation is formed using **beams of electrons** instead of light radiation.
- According to quantum mechanics, a high-velocity electron becomes wavelike, having a wavelength that is **inversely proportional** to its velocity.
- **High magnifications** and **resolving powers** of these microscopes are consequences of the short wavelengths of electron beams.
- The **electron beam** is focused and the image formed with magnetic lenses.

Microscopic Techniques

Electron Microscopy

- Both **transmission** and **reflection** beam modes of operation are possible for electron microscopes.
- *Transmission Electron Microscopy (TEM)*:
 - The image seen with a **TEM** is formed by an electron beam that passes through the specimen.
 - The image are produced by **differences** in beam scattering or diffraction produced.
 - The **transmitted beam** is projected onto a *fluorescent screen* or a photographic film so that the image may be viewed.
 - **Magnifications** with **TEM** is up to 1,000,000x.

Microscopic Techniques

Electron Microscopy

■ *Scanning Electron Microscopy (SEM):*

- A more recent and extremely useful investigative tool is the SEM.
- The surface of a specimen to be examined is scanned with an electron beam, and the reflected (*back-scattered*) beam of electrons is collected.
- Then displayed at the same scanning rate on a cathode ray tube (CRT).
- The surface may or may not be polished and etched, but it must be electrically conductive.

Microscopic Techniques

Scanning Probe Microscopy

- The field of microscopy has experienced a revolution with the development of a new family of **scanning probe microscopes**.
- The **scanning probe microscope** (SPM), differs from optical and electron microscopes in that neither **light** nor **electrons** are used to form an image.
- The **microscope generates** a ***topographical*** map on an atomic scale.
- That ***topographical map*** is a representation of surface features and characteristics of the specimen being examined.

Microscopic Techniques

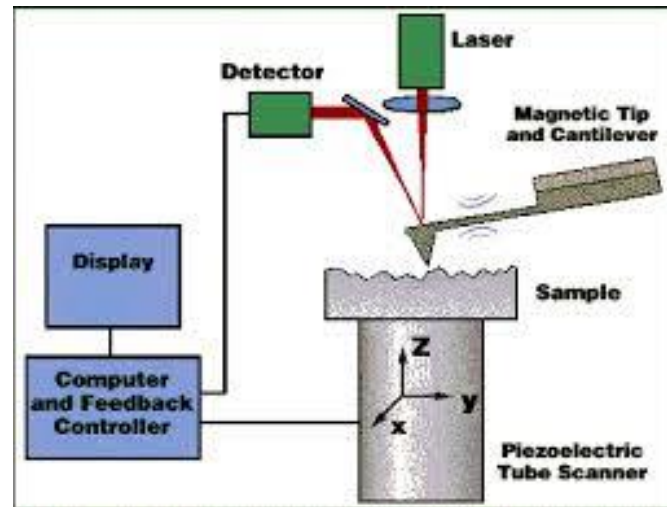
Scanning Probe Microscopy

- Some of the features that **differentiate** the **SPM** from other microscopic techniques are as follows:
 - **Better resolutions** are attainable than other microscopic techniques. (magnifications as high as 10^9)
 - **Three-dimensional** magnified images are **generated** that provide topographical information about features of interest.
 - Some **SPMs** may be operated in a *variety of environments* (e.g., **vacuum**, **air**, **liquid**). Thus a particular specimen may be examined in its most suitable environment.
- **SPMs** employ a tiny probe with a **very sharp tip** that is brought into very close proximity (a nanometer) of the specimen surface.

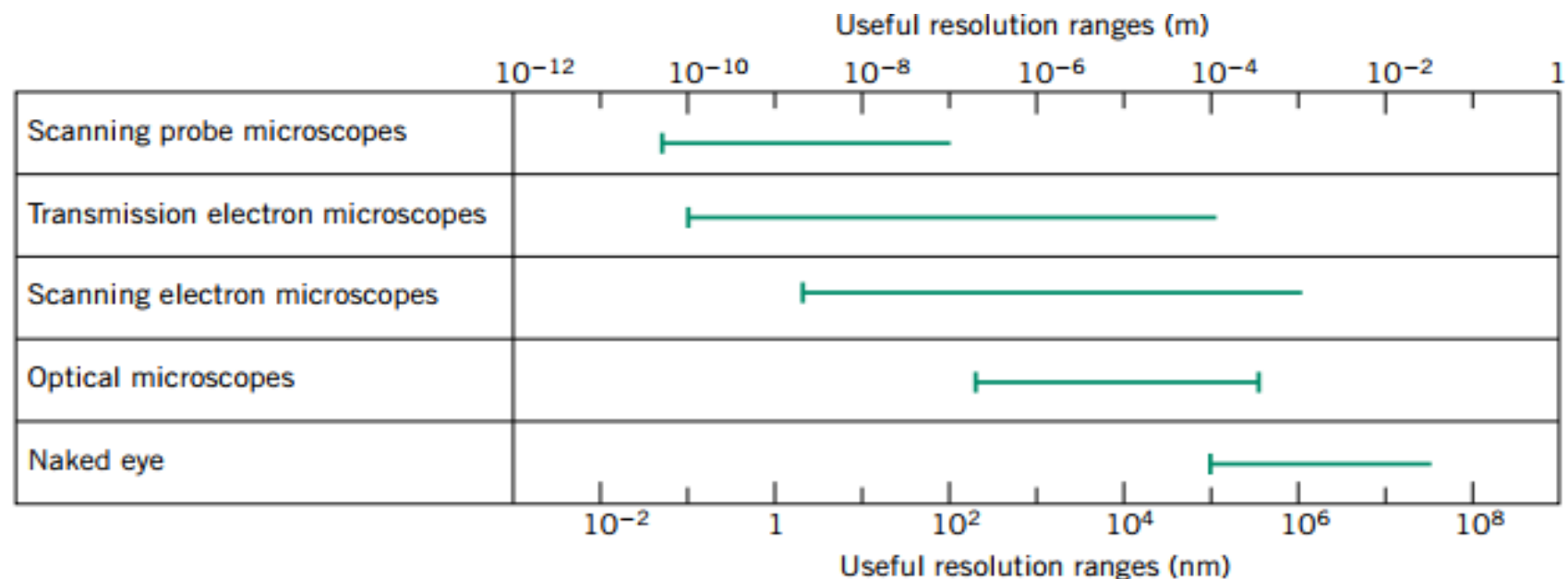
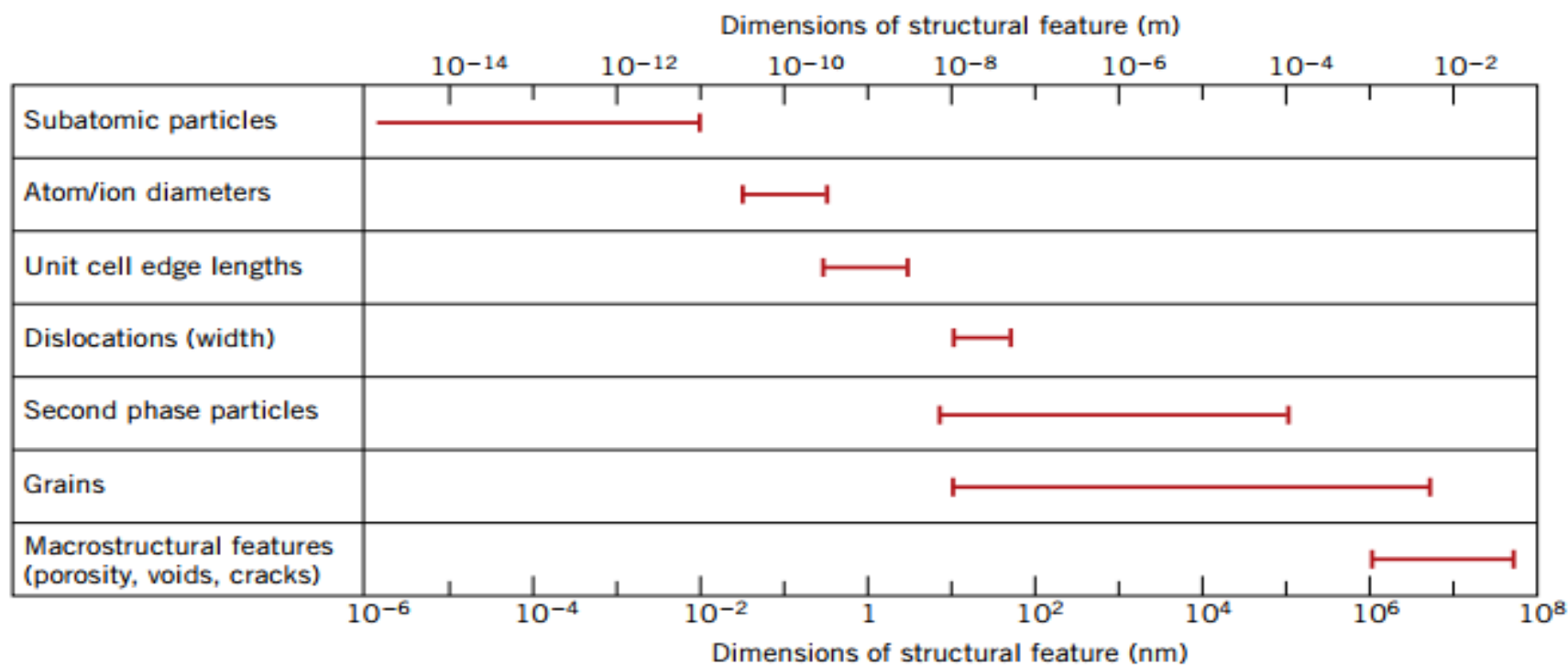
Microscopic Techniques

Scanning Probe Microscopy

- The **in-surface-plane** and **out-of-plane** motions of the probe are controlled by **piezoelectric ceramic components** that have nanometer resolutions.



Figures of the microscopes



Grain-Size Determination

- The **grain size** is often determined when the properties of polycrystalline and single-phase materials are under consideration.
- It is important to **realize** the constituent grains of materials have a variety of **shapes** and a distribution of **sizes**.
- **Grain size** may be specified in terms of average or mean grain diameter, and a number of techniques have been developed to measure this parameter.
- Historically, **grain-size** determinations were performed **manually** using **photomicrographs**.
- However, today, **most techniques** are **automated** and use digital images and image analyzers with the capacity to record, detect, and measure accurately features of the grain structure.

Grain-Size Determination

- Two common grain-size determination techniques:

1) *Linear intercept*: counting numbers of grain boundary intersections by straight test lines.

- Lines are drawn randomly through several photomicrographs that show the grain structure.
- Grain boundaries intersected by all the line segments are counted. A measure of grain diameter, may be determined

by;

$$\bar{\ell} = \frac{L_T}{PM}$$

Where, P = the sum of the total number of intersections, L_T = the total length of all the lines, M = the magnification and $\bar{\ell}$ = the mean intercept length.

2) *Comparison*: comparing grain structures with standardized charts.

Grain-Size Determination

- The **comparison method** of **grain-size determination** was devised by the American Society for Testing and Materials (ASTM).
- The ASTM has prepared several standard comparison charts, all having **different average grain sizes** and referenced to photomicrographs taken at a magnification of 100x.
- To each chart is assigned a number ranging from **1 to 10**, which is termed the ***grain-size number***.
- A **specimen** must be prepared properly to **reveal the grain structure**, which is then photographed.

Grain-Size Determination

- **Grain size** is expressed as the grain-size number of the chart that most nearly matches the grains in the micrograph.
- A relatively simple and convenient visual determination of grain-size number is possible.
- Grain-size number is used extensively in the specification of steels.
- Relationship between ASTM grain size number and number of grains per square inch (at 100x);

Visual Charts (@100x) each with a number
Quick and easy – used for steel

$$n = 2^{G-1}$$

G = Grain size number.

n = the average number of grains/square inch at 100x.

NOTE: The ASTM grain size is related (or relates) a grain area *AT 100x MAGNIFICATION*

Grain-Size Determination

- Example: For this same material, how many Grains would I expect /in² at 100x? At 50x?

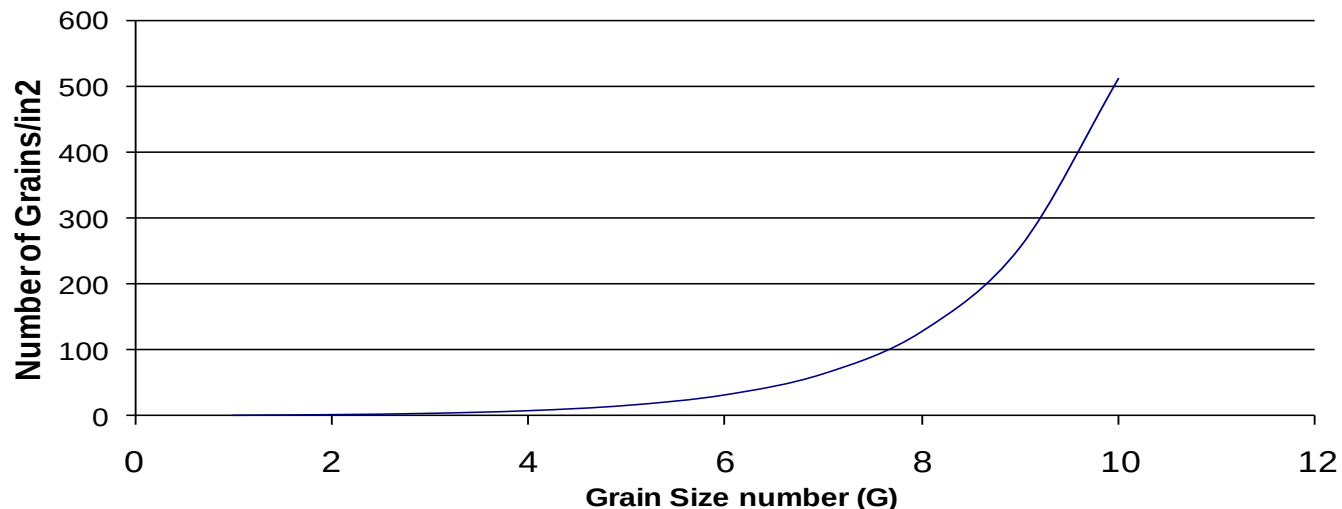
$$N = 2^{G-1} = 2^{8-1} \cong 128 \text{ grains/in}^2 \text{ (at 100x)}$$

Now, how many grain would I expect at 50x?

$$N_M = 2^{8-1} * \left(\frac{100}{M}\right)^2 = 128 * \left(\frac{100}{50}\right)^2$$

$$N_M = 128 * 2^2 = 512 \text{ grains/in}^2$$

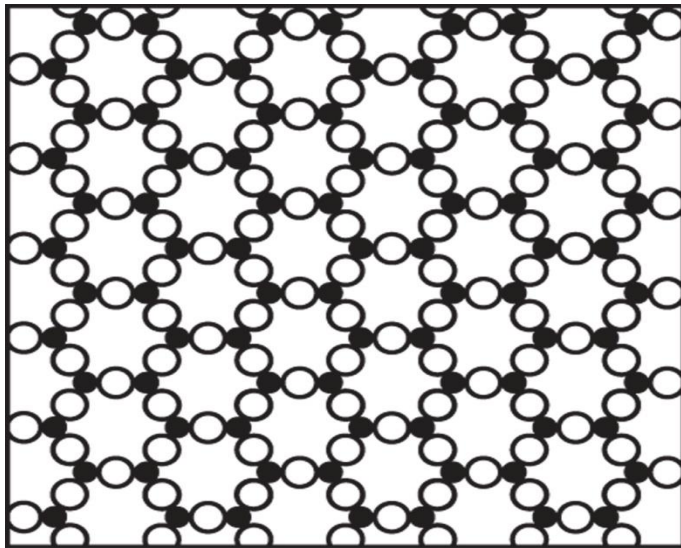
At 100x



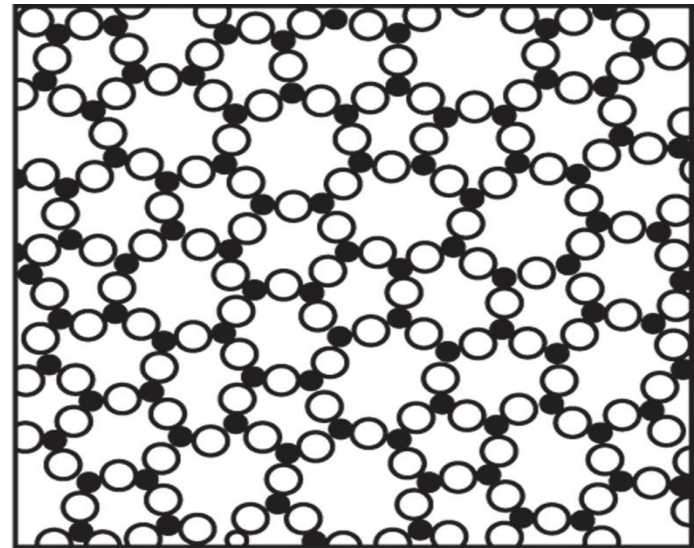
Grain-Size Determination

- Two-dimensional schematics give a comparison of (a) a *crystalline* oxide and (b) a *non-crystalline* oxide.

The *non-crystalline* material retains short-range order (the triangularly coordinated building block), but loses long-range order (*crystallinity*).



(a)



(b)

Summary

- Point, Line, Surface and Volumetric defects exist in solids.
- The number and type of defects can be varied and controlled
 - T controls vacancy conc.
 - amount of plastic deformation controls number of dislocations
 - Weight of charge materials determine concentration of substitutional or interstitial point ‘defects’
- Defects affect material properties (e.g., grain boundaries control crystal slip).
- Defects may be desirable or undesirable

Summary

- *List and explain the types of imperfections.*
- *Defects may be desirable or undesirable:*
 - *e.g., **dislocations** may be good or bad, depending on whether plastic deformation is desirable or not.*
 - *Inclusions can be intention for alloy development*
- *Point defects are those associated with one or two atomic positions.*
- ***Dislocations** are one-dimensional crystalline defects. There are two pure types: the edge and screw.*
- *An **edge** may be thought of in terms of the lattice distortion along the end of an extra half-plane of atoms.*
- *A **screw** is as a helical planar ramp.*

From W. D. Callister, 9th ed.

*Please exercise yourself with the even-question numbers by group
(Q#Even)*

End of the chapter!

Thank you!